# Millikan's oil drop experiment

The aim of the experiment was to **determine the charge** on the **electron** using a pair of parallel plates and an oil drop. It is evidence for the quantisation of electric charge.

1. A **charged** (through friction or ionisation) oil drop is allowed to fall at **terminal speed** through a chamber with **no electric field**. The **net resultant force** on the drop is **zero**. This resultant force is made up of a **drag force**, a **buoyancy force** and the **weight** of the drop. This allows the weight of the drop to be estimated.
2. The electric field is activated and used to hold the drop stationary in the chamber such that the **electric force and the weight** are **equal and opposite**. This allows the electric force on the drop to be estimated and hence its excess charge calculated.
3. After Millikan obtained the value of the **elementary charge**, he famously remarked: 'lmao imma take dis number and terrorise some G12 students lolololol'.